



CHALLENGE 19 / 20

THE ENTERPRISE-WIDE FALSE ALARM

LARGE-SCALE ALERT TRIAGE AND HIDDEN-INCIDENT DETECTION

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

A retail chain with 2,800 stores and 41,000 endpoints (point-of-sale, back-office, kiosks and servers) received 38,000 endpoint alerts for credential access within eleven minutes at 09:40 on a Monday. At 06:00 the security vendor had pushed a detection content update. The same morning there was a scheduled patch cycle, a payments software release and a new inventory agent.

Inside the surge a few things look rare: three servers contacting an unfamiliar host, and one administrator account authenticating to 14 stores in four minutes. Twelve stores are mid-festival-sale, and closing point of sale costs about ₹9 crore an hour. The console is falling behind, telemetry is delayed, and analysts see about forty alerts a minute.

Management wants to know whether to act.

2 | PROBLEM STATEMENT

Develop a **Mass Alert Validation System** that, within 15 minutes of the surge and on delayed, incomplete telemetry, decides whether it is a benign common cause, a genuine campaign or a benign cause masking a real intrusion.

The system must identify the common cause when benign, isolate the few events that do not fit it, and recommend a proportionate response with cost and risk reasoning.

Hard constraints

- The decision is due in 15 minutes and telemetry is delayed and partial.
- Several benign changes happened at once, so the cause itself is ambiguous.
- The adversary times activity to land inside the surge.
- The system must at least handle 40,000 endpoints.
- It must guard against anchoring on the benign explanation and show that it does.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Surge Characterisation	Ability to characterise a surge from partial, delayed telemetry.
02	Common-Cause Attribution	Ability to attribute a surge to one or several concurrent benign changes.
03	Campaign Detection	Ability to recognise a genuine coordinated attack.
04	Masked-Intrusion Detection	Ability to find an intrusion hidden in the noise.
05	Outlier Isolation	Ability to isolate the events that do not fit the benign explanation.
06	Cost-Aware Response Selection	Ability to weigh operational cost against risk.
07	Anchoring Resistance	Ability to avoid settling on the first plausible explanation.
08	Executive Communication	Ability to explain the decision to non-technical leadership.

The system must clearly distinguish between:

BENIGN	TRUE CAMPAIGN	MASKED INTRUSION	INCONCLUSIVE	OUTLIER
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A benign explanation must not be accepted until the events that do not fit it are examined.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated enterprise at the required scale
02	Concurrent benign causes
03	A benign-only surge, a true campaign and a masked intrusion
04	Decision within the 15-minute limit
05	Isolated outlier events
06	Cost and risk reasoning
07	Explanation to leadership

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.